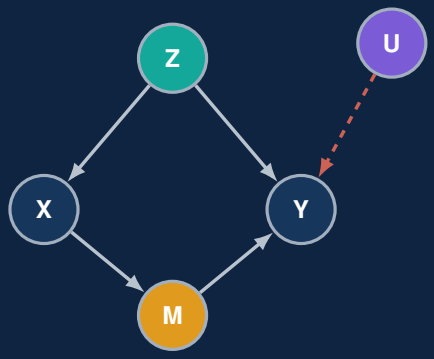


ADVANCED ARTIFICIAL INTELLIGENCE

Probabilistic Graphical Models and Causal Inference

Session 11

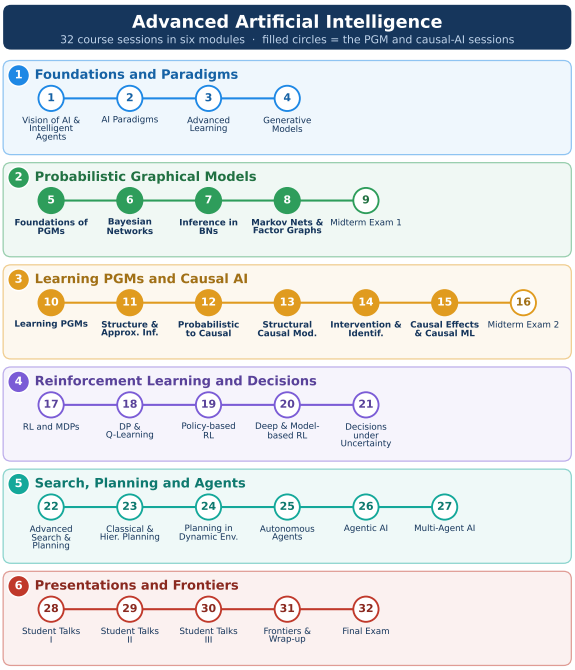
Structure Learning and Approximate Inference

Lecture Notes

Instructor: Prof. M. Arashi

About these notes

These notes cover Session 11 of the course *Advanced Artificial Intelligence*, part of its probabilistic-graphical-model and causal-AI block (course sessions 5–15). The session follows a fixed structure: a roadmap and learning objectives, the core theory organized in parts, worked numerical examples, and a part connecting the topic to intelligent agents, agentic AI, reinforcement learning, multi-agent systems and the other sessions of the course. Every number is reproduced by the companion Python scripts (`python/s11_examples.py` for the numbers, `python/s11_figures.py` for the figures), and `questions/s11_questions.tex` contains ten questions with complete answers.



The Advanced Artificial Intelligence course. Filled circles mark the PGM and causal-AI sessions; this is session 11.

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SESSION 11 ADVANCED ARTIFICIAL INTELLIGENCE

Structure Learning and Approximate Inference

Model selection, Monte Carlo sampling, MCMC and approximate inference

Advanced Artificial Intelligence · Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

PART I
Structure Learning
Scores, model selection, greedy search, the PC algorithm and CPDAGs.

PART II
Monte Carlo
Ancestral, rejection and importance sampling, likelihood weighting, ESS.

PART III
MCMC and Beyond
Gibbs, Metropolis–Hastings, diagnostics, variational inference, particle filters.

PART IV
AI Connections
MCTS, belief tracking, self-consistency, world models, agent graphs.

LEARNING OBJECTIVES

- Explain why likelihood prefers the complete DAG; compute AIC, BIC, Bayesian (BDe/K2) and held-out scores, search by hill climbing, and run the PC algorithm to a CPDAG.
- Build Monte Carlo estimators (ancestral, rejection, likelihood weighting, importance sampling), quantify their error with the CLT, and diagnose weight degeneracy with the effective sample size.
- Derive the Metropolis–Hastings acceptance ratio, prove that detailed balance implies stationarity, and read MCMC output critically (burn-in, autocorrelation, ESS, $\hat{R}$).
- Set up mean-field VI and a particle filter, and explain the asymmetry of the KL divergence.
- Connect sampling and structure learning to MCTS, belief tracking, LLM self-consistency, world models and multi-agent graphs.

Where we are. Sessions 5–8 (course sessions 5–8) built Bayesian and Markov networks and exact inference; Session 10 learned the *parameters* of a fixed graph. Today we learn the *graph* itself, and replace exact inference by sampling and optimisation when treewidth makes it hopeless, closing the probabilistic half of the notes. Session 12 opens the causal half by asking what a learned graph can and cannot tell us about interventions; sampling returns in RL (course sessions 17–20), planning (course session 22) and belief tracking (course session 21).

OPENING MOTIVATION

Two loose threads remain from Session 10. First, every session so far assumed the graph itself — who is a parent of whom, which pairs of variables share an edge — was simply handed to us by a domain expert. What if it isn't? Second, the EM algorithm's E-step leaned entirely on Session 7's variable elimination, but VE's cost is exponential in treewidth — what happens in the many realistic models (dense social networks, image grids, large medical ontologies) where treewidth is too large for exact inference to run in any reasonable time? This session closes both threads: briefly, for structure; at length, via sampling, for inference. The sampling ideas developed here — Monte Carlo error analysis and, above all, importance weighting — will return in Session 15, where inverse propensity weighting turns out to be importance sampling and Monte Carlo/bootstrap error bars quantify the uncertainty of causal effect estimates.

PART I

Learning the Graph from Data

Scores that resist overfitting, searches that scale, and tests that find v-structures

PART I · STRUCTURE LEARNING

11.1 Why Structure Learning Is Hard

Proposition 11.1 (The DAG space explodes)**.** *The number $a(n)$ of distinct DAGs on n labeled nodes grows super-exponentially in n (Robinson's recurrence $a(n) = \sum_{k=1}^{n}(-1)^{k+1}\binom{n}{k}2^{k(n-k)}a(n-k)$, $a(0)=1$, an inclusion–exclusion over the sets of source nodes); already at $n=8$ nodes there are $a(8) = 783{,}702{,}329{,}343 > 7.8\times10^{11}$ possible DAGs, compared with $2^{\binom{8}{2}} = 2^{28} \approx 2.7\times10^{8}$ undirected graphs (`ex_dag_count()`: $a(1),\dots,a(5) = 1, 3, 25, 543, 29281$). More precisely, $2^{n(n-1)/2} \le a(n) \le n!\,2^{n(n-1)/2}$, so $\log a(n)$ grows like n^2.*

Proof. Fix an ordering of the nodes. Each of the $\binom{n}{2} = n(n-1)/2$ pairs may or may not carry an edge pointing forward in that ordering, and every such choice is acyclic: this gives $2^{n(n-1)/2}$ distinct DAGs, the lower bound. Every DAG has at least one topological ordering, so every DAG is counted among the $n!$ orderings times $2^{n(n-1)/2}$ edge choices, the upper bound. For $n=8$: $2^{28} = 268{,}435{,}456 \le a(8) \le 8!\,2^{28} \approx 1.08\times10^{13}$. □

Two broad families of practical structure-learning methods exist:

- **Score-based search**: define a score (e.g., BIC, which penalizes the log-likelihood by $\frac{\log N}{2}$ times the number of free parameters, N being the number of records; defined in (11.2) directly generalizing the MLE machinery of Session 10) and search the DAG space greedily (add/remove/reverse one edge at a time) rather than exhaustively.
- **Constraint-based search**: run a sequence of conditional-independence tests (the sample analogue of the definition of conditional independence) to infer which edges *must* be absent, then orient the remaining edges using the collider-detection logic from Session 6 (a v-structure

$X \to Z \leftarrow Y$ is distinguishable from a chain precisely because X and Y test as marginally independent but as dependent given Z, whereas in a chain or fork the pattern is reversed).

Finding the highest-scoring DAG is NP-hard even when every node may have at most two parents (Chickering 1996), so both families rely on heuristics or on assumptions. In practice, on real applied problems, structure is very often *not* learned purely from data at all: an expert-elicited graph (like our student network or a clinical DAG built from established medical knowledge) is preferred whenever available, both because data-driven structure search is statistically unreliable at modest sample sizes and — as Session 12 will make sharply clear — because a graph learned purely from *observational, correlational* structure can be silent or actively misleading about which edges represent genuine causal mechanisms versus mere statistical association. We will not resolve that tension today; it is the very question the second half of these notes (Sessions 12–15) is devoted to.

PART I · STRUCTURE LEARNING

11.2 Scoring Structures: Model Selection

Fix a DAG $\mathcal{G}$ over n discrete variables, with r_i values for X_i and q_i configurations of its parents $U_i = \mathrm{pa}(X_i)$. With complete data $\mathcal{D}$ of N records, let $N_{x,u}$ be the number of records with $X_i = x$ and $U_i = u$ and $N_u = \sum_x N_{x,u}$; Session 10 gave the MLE $\hat\theta_{x|u} = N_{x,u}/N_u$. Throughout, log and ln both denote the natural logarithm.

Proposition 11.2 (Likelihood = mutual information)**.** *The maximised log-likelihood decomposes over families as*

$$\ell(\hat\theta_{\mathcal{G}} : \mathcal{D}) = N \sum_i \hat I(X_i; U_i) - N \sum_i \hat H(X_i), \tag{11.1}$$

where, for the empirical distribution $\hat P$, $\hat H(X) = -\sum_x \hat P(x) \log \hat P(x)$ is the entropy and $\hat I(X;U) = \sum_{x,u} \hat P(x,u) \log \frac{\hat P(x,u)}{\hat P(x)\hat P(u)} = \hat H(X) - \hat H(X \mid U)$ the mutual information (the KL divergence from $\hat P(x,u)$ to $\hat P(x)\hat P(u)$, which is ≥ 0 with equality iff X and U are independent under $\hat P$; its conditional version $\hat I(X; Z \mid U)$ is zero iff $X \perp\!\!\!\perp Z \mid U$). Consequently adding a parent never decreases the likelihood, and the complete DAG always maximises it.

Proof. $\ell = \sum_i \sum_{x,u} N_{x,u} \log \hat P(x \mid u) = -N \sum_i \hat H(X_i \mid U_i)$, and $\hat H(X_i \mid U_i) = \hat H(X_i) - \hat I(X_i; U_i)$. By the chain rule of mutual information, $\hat I(X; U \cup \{Z\}) = \hat I(X;U) + \hat I(X; Z \mid U) \geq \hat I(X;U)$, because conditional mutual information is a (conditional) KL divergence, hence non-negative. □

Empirical mutual information is almost never 0, so likelihood adds every edge: it *overfits*. Practical scores add a counterweight: AIC and BIC penalise the number of parameters, the Bayesian (BDe, K2) score penalises implicitly through the prior, and a held-out log-likelihood estimates predictive performance directly. Writing $d = \dim(\mathcal{G}) = \sum_i (r_i - 1) q_i$ for the number of free parameters (the parameter count of Sessions 5–6; for binary variables a node with k parents contributes 2^k):

$$\mathrm{AIC}(\mathcal{G}) = \ell(\hat\theta_{\mathcal{G}}) - d, \qquad \mathrm{BIC}(\mathcal{G}) = \ell(\hat\theta_{\mathcal{G}}) - \tfrac{\log N}{2}\, d. \tag{11.2}$$

The *Bayesian score* is the log marginal likelihood $\log P(\mathcal{D} \mid \mathcal{G}) = \log \int P(\mathcal{D} \mid \theta, \mathcal{G})\, p(\theta \mid \mathcal{G})\, d\theta$. With independent Dirichlet priors $\mathrm{Dir}(\alpha_{1|u}, \ldots, \alpha_{r_i|u})$ on every CPT row (global and local parameter

independence, Session 10) the integral factorises in closed form (Heckerman, Geiger & Chickering 1995):

$$P(\mathcal{D} \mid \mathcal{G}) = \prod_i \prod_u \frac{\Gamma(\alpha_u)}{\Gamma(\alpha_u + N_u)} \prod_x \frac{\Gamma(\alpha_{x|u} + N_{x,u})}{\Gamma(\alpha_{x|u})}, \qquad \alpha_u = \textstyle\sum_x \alpha_{x|u}. \tag{11.3}$$

All $\alpha_{x|u} = 1$ gives the *K2* score; $\alpha_{x|u} = \alpha/(r_i q_i)$ gives *BDeu* with equivalent sample size α. Unlike the likelihood, (11.3) penalises complexity automatically: a richer graph spreads its prior over more parameters (the Bayesian Occam's razor). Finally, *cross-validation* scores a structure by the log-likelihood of held-out data; per record this estimates $\mathbb{E}_{P^*} \log \hat{P}_{\mathcal{G}}(\mathbf{X}) = -H(P^*) - \mathrm{KL}(P^* \| \hat{P}_{\mathcal{G}})$, i.e. minus the predictive KL divergence up to the constant entropy $H(P^*)$ of the true distribution P^*.

The scores at work. Take the binary student network of Worked Example 11.2 ($D \to G \leftarrow I$, $I \to S$, $G \to L$). From $N = 300$ training records we fit a nested sequence of DAGs that first adds the four true edges and then five spurious ones, and evaluate each on 20,000 fresh test records. The training log-likelihood per record rises monotonically, from -3.2560 (empty) through -2.5683 (true graph) to -2.5393 (all nine edges), as Proposition 11.2 guarantees; the test log-likelihood, BIC and AIC all peak at the true graph.

PYTHON LAB · `s11_examples.py` · `ex_score_curve()`

```
edges=0 k= 5 train=-3.2560 test=-3.3212 | edges=4 k=10 train=-2.5683
test=-2.6317 BIC/N=-2.6633 AIC/N=-2.6016
edges=9 k=30 train=-2.5393 test=-2.6526   BIC picks 4, AIC picks 4, test-LL
best at 4
```

For *nested* structures $\mathcal{G}_1 \subset \mathcal{G}_2$ the classical alternative is the likelihood-ratio test: under $\mathcal{G}_1$, $2(\ell_2 - \ell_1) \approx \chi^2_{d_2 - d_1}$ for large N (Wilks' theorem), so $\mathcal{G}_1$ is rejected at level 5% when $2(\ell_2 - \ell_1)$ exceeds the 0.95 quantile (3.84 for one degree of freedom). BIC prefers $\mathcal{G}_2$ iff $2(\ell_2 - \ell_1) > (d_2 - d_1)\log N$, a threshold that grows with N; this is why BIC, unlike AIC or a fixed-level test, eventually stops adding spurious edges.

All five scores are *decomposable*: a sum of family scores $\sum_i \mathrm{FamScore}(X_i, U_i : \mathcal{D})$, each depending only on the counts of X_i and its parents. Likelihood, AIC, BIC and BDeu are also *score equivalent*: Markov-equivalent DAGs (Theorem 11.1) get the same score (they have the same maximised likelihood and the same number of parameters; K2 is the exception).

Worked Example 11.1 (A BIC calculation, structure search made concrete)**.** A structure-learning algorithm compares two candidate structures for (A, B) using $N = 100$ fully-observed records: 60 have $A{=}1$, 40 have $A{=}0$; among $A{=}1$, $B{=}1$ occurs 48 times; among $A{=}0$, $B{=}1$ occurs 8 times. Compare $\mathrm{BIC} = \ell(\hat\theta) - \frac{d}{2}\log N$ for the edge model ($A \to B$) versus the no-edge model ($A \perp B$).

Solution. **Edge model.** MLEs: $\hat P(A{=}1) = 0.6$, $\hat P(B{=}1 \mid A{=}1) = 48/60 = 0.8$, $\hat P(B{=}1 \mid A{=}0) = 8/40 = 0.2$. Log-likelihood: $\ell_{\text{edge}} = 60\ln 0.6 + 40\ln 0.4 + 48\ln 0.8 + 12\ln 0.2 + 8\ln 0.2 + 32\ln 0.8$. (the four B-counts are $N_{B=1,A=1} = 48$, $N_{B=0,A=1} = 60 - 48 = 12$, $N_{B=1,A=0} = 8$, $N_{B=0,A=0} = 40 - 8 = 32$). Grouping the B-terms ($48 + 32 = 80$ occurrences of $\ln 0.8$, $12 + 8 = 20$ of $\ln 0.2$):

$$\begin{aligned}\ell_{\text{edge}} &= 60(-0.5108) + 40(-0.9163) + 80(-0.2231) + 20(-1.6094)\\ &= -30.65 - 36.65 - 17.85 - 32.19 = -67.30 - 50.04 = \mathbf{-117.34}.\end{aligned}$$

With $d_{\text{edge}} = 3$ free parameters (one for $P(A)$, two for $P(B \mid A)$) and $\ln 100 = 4.6052$, $\text{BIC}_{\text{edge}} = -117.34 - 1.5 \cdot 4.6052 = -117.34 - 6.91 = \mathbf{-124.25}$.
No-edge model. $\hat{P}(B{=}1) = (48 + 8)/100 = 0.56$ (pooled). $\ell_{\text{indep}} = [60 \ln 0.6 + 40 \ln 0.4] + [56 \ln 0.56 + 44 \ln 0.44] = -67.30 + (-32.47 - 36.12) = -67.30 - 68.59 = \mathbf{-135.89}$. With $d_{\text{indep}} = 2$, $\text{BIC}_{\text{indep}} = -135.89 - \frac{2}{2} \cdot 4.6052 = \mathbf{-140.50}$.
Since $-124.25 > -140.50$ by a wide margin ($-124.25 - (-140.50) = 16.25$), the edge model wins decisively: the data's strong dependence (0.8 vs. 0.2) far outweighs the extra parameter's complexity penalty at $N = 100$. AIC agrees: $-117.34 - 3 = -120.34$ against $-135.89 - 2 = -137.89$. By (11.1) the likelihood gain $-117.34 - (-135.89) = 18.55$ is exactly $N\hat{I}(A; B)$. The reversed DAG $B \to A$ has the B-family $56 \ln 0.56 + 44 \ln 0.44 = -68.59$ and the $A \mid B$ family $48 \ln \frac{48}{56} + 8 \ln \frac{8}{56} + 12 \ln \frac{12}{44} + 32 \ln \frac{32}{44} = -48.75$, in total the same -117.34: data cannot orient a single edge. (An earlier edition of these notes gave -135.87 and -140.48; the values above are recomputed.)

PART I · STRUCTURE LEARNING

11.3 Score-Based Search: Greedy Hill Climbing

Decomposability is what makes search affordable: a local move changes the parent set of one node (add or delete $u \to v$) or two nodes (reverse $u \to v$), so its score change Δ needs only one or two family scores, which can be cached.

Greedy hill climbing (1) starts from the empty graph (or a random DAG, for a restart) and caches family scores; (2) evaluates every legal move (add, delete or reverse one edge, keeping acyclicity and an in-degree bound) through the one or two changed families; (3) applies the best move whose result is not on a short *tabu list* of recently visited graphs; (4) stops when no move improves the score, and (5) repeats from several random starts, returning the best local optimum. Each step examines $O(n^2)$ moves; local optima and plateaus of Markov-equivalent graphs motivate tabu lists and restarts, and searching over equivalence classes (GES) is a refinement (Koller & Friedman 2009, Ch. 18).

Worked Example 11.2 (Hill climbing recovers the student network)**.** The *student network* of Session 6, in a hypothetical all-binary version: Difficulty D (1 = hard), Intelligence I (1 = high), Grade G (1 = good), SAT S (1 = high), Letter L (1 = strong), with DAG $D \to G \leftarrow I$, $I \to S$, $G \to L$ and illustrative CPTs $P(D{=}1) = 0.4$, $P(I{=}1) = 0.3$; $P(G{=}1 \mid d, i) = 0.5, 0.9, 0.2, 0.6$ for $(d, i) = (0,0), (0,1), (1,0), (1,1)$; $P(S{=}1 \mid i) = 0.1, 0.8$ for $i = 0, 1$; $P(L{=}1 \mid g) = 0.1, 0.85$ for $g = 0, 1$. Sample $N = 5000$ records (simulation, seed 2026) and hill-climb with BIC (at most 3 parents, tabu list of the last 10 graphs) from five starts: the empty graph and four random DAGs (random node order, each forward pair joined with probability 0.3).
Solution (simulation output). From the empty graph four moves reach BIC -13003.29, which is exactly the BIC of the true DAG, with exactly its edges; restart 4 also reaches it, while restarts 1–3 stop at worse local optima (-13029.95, -13014.62, -13007.31). The v-structure is oriented because reversing one of its arrows changes the equivalence class and the score (reversing $D \to G$ gives -13031.58); the direction of $I \to S$, in contrast, is an arbitrary tie-break, since $S \to I$ is Markov equivalent and has the same BIC -13003.29.

PYTHON LAB · s11_examples.py · ex_hill_climb()

```
restart 0:  BIC = -13003.29 after 4 moves, edges (D,G),(I,G),(I,S),(G,L) =
true DAG
restarts 1-4:  -13029.95, -13014.62, -13007.31, -13003.29; S->I instead of
I->S: -13003.29; G->D: -13031.58
```

PART I · STRUCTURE LEARNING

11.4 Constraint-Based Learning: The PC Algorithm

Definition 11.1 (Markov equivalence, CPDAG)**.** Two DAGs are *Markov equivalent* if they encode the same set of conditional independences. The *CPDAG* (essential graph) of an equivalence class has a directed edge where every member agrees on the orientation and an undirected edge elsewhere.

Theorem 11.1 (Verma & Pearl 1990)**.** *Two DAGs are Markov equivalent if and only if they have the same skeleton and the same v-structures (colliders $X \to Z \leftarrow Y$ with X, Y non-adjacent).*

Because observational data only reveal independences, *no* method — score-based or constraint-based — can do better than the CPDAG from observational data alone. The PC algorithm (Spirtes, Glymour & Scheines 2000) finds it with CI tests:

PC algorithm. (1) Start from the complete undirected graph. (2) For $\ell = 0, 1, 2, \dots$ (while some node has more than ℓ neighbours): for each adjacent pair $X - Y$ and each $S \subseteq \mathrm{adj}(X) \setminus \{Y\}$ with $|S| = \ell$, if the test accepts $X \perp\!\!\!\perp Y \mid S$, delete $X - Y$ and store $\mathrm{sep}(X, Y) = S$. (3) For every non-adjacent pair X, Y with common neighbour $Z \notin \mathrm{sep}(X, Y)$, orient $X \to Z \leftarrow Y$. (4) Apply Meek's rules, e.g. R1: $X \to Z - W$ with X, W non-adjacent $\Rightarrow Z \to W$ (otherwise a new v-structure would appear).

Under *faithfulness* (every independence in P is a d-separation of the true DAG) and a perfect CI oracle, PC returns exactly the true CPDAG, in polynomial time for bounded degree. In practice the test is the likelihood-ratio G^2-test (discrete data), with statistic $G^2 = 2N\hat{I}(X; Y \mid S)$, approximately χ^2_{df} under independence with $\mathrm{df} = (r_X - 1)(r_Y - 1)\prod_{Z \in S} r_Z$ (for binary variables $\mathrm{df} = 2^{|S|}$); or, for Gaussian data, Fisher's z-test: with $\hat{r}$ the sample partial correlation of X, Y given S from n observations, $z = \frac{1}{2}\ln\frac{1+\hat{r}}{1-\hat{r}} \cdot \sqrt{n - |S| - 3}$ is compared with $N(0, 1)$ quantiles.

Worked Example 11.3 (PC on the student data)**.** Run PC with G^2-tests at level 0.01 on a fresh sample of $N = 5000$ records from the network of Worked Example 11.2 (simulation, seed 2027). A pair is declared independent when $p > 0.01$.
Solution (Fig. 11.1; simulation output). (1) Order 0 removes $D - I$ ($p = 0.476$) and $D - S$ ($p = 0.916$); every other marginal test rejects (e.g. $G^2 = 257.59$, $\mathrm{df} = 1$, $p = 5.8 \times 10^{-58}$ for D–L). (2) Order 1 removes $D - L$ ($p = 0.749$) and $I - L$ ($p = 0.335$) given G, and $G - S$ ($p = 0.183$) and $S - L$ ($p = 0.691$) given I. (3) D and I are non-adjacent with common neighbour $G \notin \mathrm{sep}(D, I) = \emptyset$, so $D \to G \leftarrow I$ — the collider signature $D \perp\!\!\!\perp I$ but $D \not\perp\!\!\!\perp I \mid G$ ($G^2 = 65.87$, $\mathrm{df} = 2$, $p = 5.0 \times 10^{-15}$). (4) Meek R1 with $D \to G - L$ and D, L non-adjacent gives $G \to L$; $I - S$ stays undirected, since no arrow points into I or S and $I \to S$, $S \to I$ are

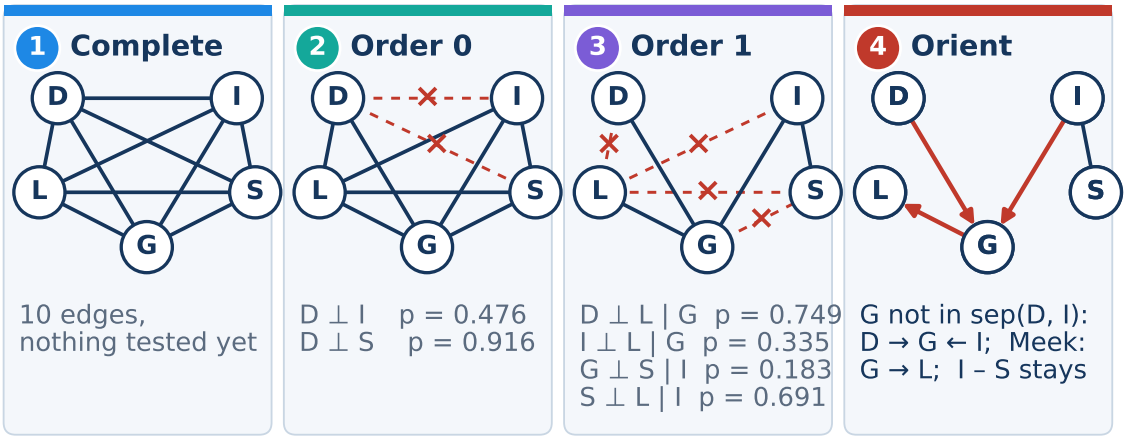


Figure 11.1. The PC algorithm on $N = 5000$ records from the student network of Worked Example 11.2. Red dashed edges are deleted at each order, with the G^2-test p-values; the final panel is the CPDAG: $D \to G \leftarrow I$ and $G \to L$ are compelled, $I - S$ cannot be oriented (`ex_pc_algorithm()`).

Markov equivalent.

PYTHON LAB · `s11_examples.py` · `ex_pc_algorithm()`

```
D _||_ I: G = 0.51, p = 0.476; D _||_ I | G: G = 65.87, df = 2, p = 4.98e-15;
D _||_ L: G = 257.59
removed D-I, D-S (order 0); D-L, I-L | G and G-S, S-L | I (order 1; p =
0.749, 0.335, 0.183, 0.691)
CPDAG directed:  (D,G),(G,L),(I,G); undirected:  (I,S)
```

PART II

Sampling-Based Inference

Replace sums by averages over random draws — and pay attention to the weights

PART II · MONTE CARLO

11.5 When Exact Inference Fails: The Case for Sampling

Recall Session 7's headline result: for n variables with at most r values each, VE costs $O(n \cdot r^{w+1})$ where w is the treewidth. Many real graphical models — densely connected social networks, image grids with strong neighbor coupling, gene regulatory networks — have treewidth comparable to n itself, making exact inference computationally hopeless regardless of how cleverly we order eliminations. *Sampling-based approximate inference* trades exactness for tractability: instead of computing $P(y \mid e)$ exactly, we draw samples from (a distribution close to) P, and estimate $P(y \mid e)$ by counting.

The Monte Carlo principle. Any query is an expectation: $P(y) = \mathbb{E}_P[\mathbb{1}\{Y = y\}]$. Given $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)} \overset{\text{iid}}{\sim} P$,

$$\hat{\mu}_N = \frac{1}{N}\sum_{m=1}^{N} f(\mathbf{x}^{(m)}) \;\approx\; \mu = \mathbb{E}_P[f(\mathbf{X})], \qquad \mathrm{SE}(\hat{\mu}_N) = \frac{\sigma_f}{\sqrt{N}}, \quad \sigma_f^2 = \mathrm{Var}_P f(\mathbf{X}), \tag{11.4}$$

whatever the dimension of $\mathbf{x}$: $\hat{\mu}_N$ is unbiased with variance σ_f^2/N, converges to μ by the law of large numbers, and by the central limit theorem $\hat{\mu}_N \pm 1.96\, s_N/\sqrt{N}$ (s_N the sample standard deviation of the $f(\mathbf{x}^{(m)})$) is an approximate 95% confidence interval. For a probability query $f = \mathbb{1}\{Y = y\}$, $\sigma_f^2 = p(1-p)$ with $p = P(y)$. The error decays like $1/\sqrt{N}$: each extra digit of accuracy costs $100\times$ more samples.

11.5.1 Forward (Ancestral) Sampling

Definition 11.2 (Ancestral sampling)**.** Given a Bayesian network with no evidence, sample every variable in an order consistent with the DAG (parents before children, always possible since the graph is acyclic): draw $x_i \sim P(X_i \mid \mathrm{pa}(X_i))$ with the parents set to their already-sampled values.

This produces exact i.i.d. samples from the joint $P(\mathbf{x})$ with no approximation at all — it is simply the BN chain rule $P(\mathbf{x}) = \prod_i P(x_i \mid \mathrm{pa}_i)$ (Session 6) turned into a generative procedure. The catch appears once we need to condition on evidence: if evidence $E = e$ is rare, most sampled trajectories will simply not match e and must be discarded, wasting nearly all the computation (*rejection sampling*). *Importance sampling* improves on this by sampling from a more convenient proposal distribution Q and reweighting each sample by $P(\mathbf{x})/Q(\mathbf{x})$ rather than discarding mismatches outright — a technique we will meet again, doing essentially the same reweighting job, as inverse propensity weighting in Session 15.

Worked Example 11.4 (Monte Carlo on the sprinkler network)**.** The sprinkler network of Russell & Norvig (2021): $C \to S$, $C \to R$, $(S, R) \to W$ with $P(c) = 0.5$, $P(s \mid c) = 0.1$, $P(s \mid \neg c) = 0.5$, $P(r \mid c) = 0.8$, $P(r \mid \neg c) = 0.2$, $P(w \mid s, r) = 0.99$, $P(w \mid s, \neg r) = P(w \mid \neg s, r) = 0.90$, $P(w \mid \neg s, \neg r) = 0$. Estimate $P(W{=}1)$ from $N = 1000$ ancestral samples.
Solution. *Exact value by enumeration.* First $P(s, r) = \sum_c P(c)P(s \mid c)P(r \mid c)$: $P(S{=}1, R{=}1) = 0.5(0.1{\cdot}0.8) + 0.5(0.5{\cdot}0.2) = 0.04 + 0.05 = 0.09$, $P(S{=}1, R{=}0) = 0.5(0.1{\cdot}0.2) + 0.5(0.5{\cdot}0.8) = 0.01 + 0.20 = 0.21$, $P(S{=}0, R{=}1) = 0.5(0.9{\cdot}0.8) + 0.5(0.5{\cdot}0.2) = 0.36 + 0.05 = 0.41$, $P(S{=}0, R{=}0) = 0.5(0.9 \cdot 0.2) + 0.5(0.5 \cdot 0.8) = 0.09 + 0.20 = 0.29$. Then

$$\begin{aligned} P(W{=}1) &= \textstyle\sum_{s,r} P(s, r)P(W{=}1 \mid s, r) = 0.09 \cdot 0.99 + 0.21 \cdot 0.90 + 0.41 \cdot 0.90 + 0.29 \cdot 0 \\ &= 0.0891 + 0.1890 + 0.3690 = 0.6471. \end{aligned}$$

Monte Carlo (one simulated run, seed 2026). The sample proportion is $\hat{p} = 0.6330$ with $\mathrm{SE} = \sqrt{\hat{p}(1-\hat{p})/N} = \sqrt{0.6330 \cdot 0.3670/1000} = 0.0152$, so the 95% interval $0.6330 \pm 1.96 \cdot 0.0152 = [0.6031, 0.6629]$ covers the truth. *Planning.* Since $\sigma^2 = 0.6471 \cdot 0.3529 = 0.2284$, $\sigma = 0.478$, the SE $\sigma/\sqrt{N}$ is 0.048, 0.0048, 0.00048 at $N = 10^2, 10^4, 10^6$; an SE of 0.005 needs $N = \sigma^2/0.005^2 = 0.2284/0.000025 = 9134.5$, i.e. $N = 9135$ samples.

PYTHON LAB · s11_examples.py · ex_mc_basics()

```
exact P(W=1) = 0.6471; N = 1000:  estimate = 0.6330, SE = 0.0152, 95% CI =
[0.6031, 0.6629]; N for SE <= 0.005:  9,135
```

PART II · MONTE CARLO

11.6 Rejection Sampling and Likelihood Weighting

Rejection sampling for $P(y \mid e)$ draws ancestral samples and keeps those with $E = e$; the kept samples are exact draws from $P(\cdot \mid e)$, but the acceptance rate is $P(e)$, so the expected number of draws per kept sample is $1/P(e)$, and $P(e)$ typically shrinks exponentially in the number of evidence variables.

Likelihood weighting (LW) never rejects: it clamps the evidence variables, samples the others ancestrally, and gives each sample the weight

$$w(\mathbf{x}) = \prod_{i:\, X_i \in E} P(e_i \mid \mathrm{pa}(X_i)), \qquad \widehat{P}(y \mid e) = \frac{\sum_m w^{(m)} \mathbb{1}\{y^{(m)} = y\}}{\sum_m w^{(m)}}. \tag{11.5}$$

LW is importance sampling (next section) with the proposal "sample non-evidence variables from their CPTs": the ratio $P(\mathbf{x}, e)/Q(\mathbf{x})$ leaves exactly the evidence CPTs. Its weakness: evidence only influences its *descendants*' sampling; upstream variables are sampled from the prior and merely re-weighted.

Worked Example 11.5 (Rejection versus likelihood weighting)**.** In the sprinkler network, estimate $P(R{=}1 \mid S{=}1, W{=}1)$ with $N = 10{,}000$ samples each way.
Solution. *Exact value*, with the $P(s, r)$ of Worked Example 11.4: $P(S{=}1, R{=}1, W{=}1) = 0.09 \cdot 0.99 = 0.0891$ and $P(S{=}1, R{=}0, W{=}1) = 0.21 \cdot 0.90 = 0.1890$, so $P(S{=}1, W{=}1) = 0.0891 + 0.1890 = 0.2781$ and $P(R{=}1 \mid S{=}1, W{=}1) = 0.0891/0.2781 = 0.3204$. The sprinkler explains away the wet grass: without $S{=}1$, $P(R{=}1, W{=}1) = 0.0891 + 0.41 \cdot 0.90 = 0.4581$ and $P(R{=}1 \mid W{=}1) = 0.4581/0.6471 = 0.7079$. *Simulation* (seed 2026). Rejection keeps 2852 of 10,000 samples (28.5%, close to $P(S{=}1, W{=}1) = 27.8\%$), estimate 0.3310. LW samples C and R from their CPTs and weights each sample by $P(S{=}1 \mid c)\, P(W{=}1 \mid S{=}1, r)$: $0.1 \cdot 0.99 = 0.099$, $0.1 \cdot 0.90 = 0.090$, $0.5 \cdot 0.99 = 0.495$, $0.5 \cdot 0.90 = 0.450$ for $(C, R) = (1,1), (1,0), (0,1), (0,0)$. All samples are used, the estimate is 0.3301, and the effective sample size $\mathrm{ESS} = (\sum w)^2 / \sum w^2$ ((11.7), next section) is 7012.

PYTHON LAB · s11_examples.py · ex_rejection_lw()

```
exact P(R=1 | S=1, W=1) = 0.3204; rejection:  2852 of 10000 kept (28.5%),
estimate 0.3310
likelihood weighting:  estimate = 0.3301; ESS = 7012 (70.1%)
```

PART II · MONTE CARLO

11.7 Importance Sampling and the Effective Sample Size

For any proposal q with $q(x) > 0$ wherever $f(x)p(x) \neq 0$,

$$\mathbb{E}_p[f(X)] = \mathbb{E}_q\big[w(X)f(X)\big], \quad w = \frac{p}{q}; \qquad \hat{\mu}_{\mathrm{IS}} = \frac{1}{N}\sum_m w^{(m)} f^{(m)}, \quad \hat{\mu}_{\mathrm{SNIS}} = \frac{\sum_m w^{(m)} f^{(m)}}{\sum_m w^{(m)}}. \tag{11.6}$$

The self-normalised form only needs p up to a constant (posteriors, Markov networks) and is consistent but slightly biased. The variance of $\hat{\mu}_{\mathrm{IS}}$ is governed by $\mathbb{E}_q[w^2 f^2]$; a heavy-tailed weight distribution means a few samples dominate. The standard diagnostic is

$$\mathrm{ESS} = \frac{\big(\sum_m w^{(m)}\big)^2}{\sum_m \big(w^{(m)}\big)^2} \in [1, N], \tag{11.7}$$

the number of equally weighted samples with the same variance (for f that is roughly constant); for $N \to \infty$, $\mathrm{ESS}/N \to 1/\mathbb{E}_q[w^2]$ when $\mathbb{E}_q w = 1$.

Worked Example 11.6 (Self-normalised IS and the ESS by hand)**.** Five draws from a proposal have importance weights $w = (0.5,\ 2.5,\ 1.0,\ 0.25,\ 0.75)$ and function values $f = (3,\ 1,\ 2,\ 6,\ 2)$. Compute $\hat{\mu}_{\mathrm{IS}}$, $\hat{\mu}_{\mathrm{SNIS}}$ and the ESS.
Solution. $\sum w = 0.5 + 2.5 + 1.0 + 0.25 + 0.75 = 5.00$; $\sum wf = 1.5 + 2.5 + 2.0 + 1.5 + 1.5 = 9.00$; $\sum w^2 = 0.25 + 6.25 + 1 + 0.0625 + 0.5625 = 8.125$. Hence $\hat{\mu}_{\mathrm{IS}} = 9.00/5 = 1.800$ and $\hat{\mu}_{\mathrm{SNIS}} = 9.00/5.00 = 1.800$ (they coincide only because these weights happen to average exactly 1), and $\mathrm{ESS} = 5.00^2/8.125 = 3.077$ of $n = 5$. The normalised weights $(0.10, 0.50, 0.20, 0.05, 0.15)$ show why: half of the total weight sits on one draw.

PYTHON LAB · `s11_examples.py` · `ex_ess_hand()`

```
sum w = 5.00; sum w f = 9.00; sum w^2 = 8.1250; IS = 1.800; SNIS = 1.8000;
ESS = 3.077 of n = 5
```

PART III

MCMC, Variational Inference and Particle Filters

Walk instead of jump; optimise instead of sample; filter instead of smooth

PART III · MCMC AND BEYOND

11.8 Markov Chains and Gibbs Sampling

When even constructing a good proposal Q is hard (common for densely-connected Markov networks, where there is no natural "ancestral" order at all), we turn to Markov chain Monte Carlo (MCMC): build a Markov chain over the state space whose *stationary distribution* is exactly the target P, then run it long enough and treat the states visited as (correlated, but eventually representative) samples from P.

Definition 11.3 (Stationarity and detailed balance)**.** A transition kernel $T(\mathbf{x} \to \mathbf{x}')$ has *stationary distribution* π if $\sum_{\mathbf{x}} \pi(\mathbf{x}) T(\mathbf{x} \to \mathbf{x}') = \pi(\mathbf{x}')$ for all $\mathbf{x}'$. It satisfies *detailed balance* (is *reversible*) with respect to π if $\pi(\mathbf{x}) T(\mathbf{x} \to \mathbf{x}') = \pi(\mathbf{x}') T(\mathbf{x}' \to \mathbf{x})$ for all pairs.

Detailed balance implies stationarity: summing $\pi(\mathbf{x}) T(\mathbf{x} \to \mathbf{x}') = \pi(\mathbf{x}') T(\mathbf{x}' \to \mathbf{x})$ over $\mathbf{x}$ gives $\sum_{\mathbf{x}} \pi(\mathbf{x}) T(\mathbf{x} \to \mathbf{x}') = \pi(\mathbf{x}')$, because each row of T sums to one. For a finite irreducible, aperiodic chain the stationary distribution is unique and the chain converges to it from any start, and time averages along the chain converge to expectations under π (the ergodic theorem).

Algorithm 11.1 Gibbs Sampling

1: Initialize $\mathbf{x}^{(0)}$ arbitrarily.
2: **for** $t = 0, 1, 2, \ldots$ **do**
3: **for** $i = 1, \ldots, n$ (a fixed order; or pick i at random) **do**
4: Sample $x_i^{(t+1)} \sim P\big(X_i \mid x_1^{(t+1)}, \ldots, x_{i-1}^{(t+1)}, x_{i+1}^{(t)}, \ldots, x_n^{(t)}\big)$, the full conditional given the *current* values of all other variables (those already updated in this sweep at their new values).
5: **end for**
6: **end for**

The full conditional $P(X_i \mid x_{-i})$ has a beautiful simplification in a Markov network (Session 8): it only depends on X_i's *neighbors* in the graph, since every potential not involving X_i cancels in the normalisation — the neighbors form X_i's Markov blanket. This is another direct payoff of the undirected graphical structure; in a Bayesian network the blanket is parents, children and co-parents (Session 6), and $P(x_i \mid x_{-i}) \propto P(x_i \mid \mathrm{pa}_i) \prod_{j \in \mathrm{ch}(i)} P(x_j \mid \mathrm{pa}_j)$.

Proposition 11.3 (Gibbs sampling is correct, informally)**.** *The Markov chain defined by Gibbs sampling has the target distribution P as a stationary distribution: if $\mathbf{x}^{(t)} \sim P$, then $\mathbf{x}^{(t+1)} \sim P$ as well. If $P(\mathbf{x}) > 0$ for every state, it is the unique stationary distribution.*

The proof is a direct check of *detailed balance* for one coordinate update: with $\mathbf{x}_{-i} = \mathbf{x}'_{-i}$, $P(\mathbf{x}) T(\mathbf{x} \to \mathbf{x}') = P(\mathbf{x}_{-i}) P(x_i \mid \mathbf{x}_{-i}) P(x'_i \mid \mathbf{x}_{-i})$ is symmetric in $x_i \leftrightarrow x'_i$, i.e. in stationarity the chain is exactly as likely to move from state a to state b as from b to a, which is a sufficient condition for P to be stationary for that update; a sweep is a composition of such updates, each of which leaves P unchanged. Uniqueness needs every state to be reachable, which holds when all potentials are strictly positive. In practice, the chain needs some number of iterations ("burn-in") before it forgets its (usually poor) initialization and starts producing draws representative of P.

INTUITION

Do not confuse the number of Gibbs sweeps with the number of independent samples: consecutive draws are correlated (each differs from the last by only one variable's resampling). Practitioners discard a burn-in period and measure the correlation through the effective sample size (Definition 11.4). Thinning the chain (keeping only every k-th sample) makes the kept draws closer to independent, but it throws information away: it saves memory and never increases the ESS compared with keeping all draws (Section 11.10).

Worked Example 11.7 (Gibbs sampling a small Markov network, exactly checked)**.** Consider three binary spins $X_1, X_2, X_3 \in \{-1, +1\}$ arranged in a triangle (every pair connected), with pairwise potentials $\phi(x_i, x_j) = \exp(J\, x_i x_j)$, $J = 0.9$, favoring agreement (the same

style of model as Session 8's grid, small enough here that we can also compute the exact answer by brute-force enumeration over all $2^3 = 8$ states, for comparison). *Exact distribution.* $P(\mathbf{x}) = e^{J(x_1x_2+x_2x_3+x_1x_3)}/Z$. In an all-agreeing state the three products are $+1$, giving weight $e^{3J} = e^{2.7} = 14.880$; in each of the six mixed states one spin disagrees with the other two, so the products sum to -1 and the weight is $e^{-J} = e^{-0.9} = 0.4066$. Hence $Z = 2 \cdot 14.880 + 6 \cdot 0.4066 = 32.199$, $P(+,+,+) = P(-,-,-) = 14.880/32.199 = 0.4621$ and each mixed state has $0.4066/32.199 = 0.0126$. *Simulation* (Algorithm 11.1 with systematic scan, start $(+1,-1,+1)$, seed 7). We ran 3000 sweeps and discarded the first 50 as burn-in. The empirical frequencies of the remaining 2950 sweeps (0.4922 for $(+,+,+)$ and 0.4261 for $(-,-,-)$, every mixed state within 0.003 of 0.0126) match the exact values to within 0.0360. The residual error is Monte Carlo error inflated by autocorrelation: the chain switches between the two dominant states only rarely. This confirms Proposition 11.3 numerically without ever computing the partition function Z directly — exactly the situation we will face in larger, intractable models where brute-force checking is no longer available.

PART III · MCMC AND BEYOND

11.9 Metropolis–Hastings

Gibbs needs full conditionals we can sample. Metropolis–Hastings (MH; Metropolis et al. 1953, Hastings 1970) needs only the ability to evaluate $\tilde{\pi} \propto \pi$ and to sample some proposal $q(\mathbf{x}' \mid \mathbf{x})$.

Deriving the acceptance ratio. Propose $\mathbf{x}' \sim q(\cdot \mid \mathbf{x})$ and accept with probability $\alpha(\mathbf{x}, \mathbf{x}')$, otherwise stay. For $\mathbf{x}' \neq \mathbf{x}$ the kernel is $T(\mathbf{x} \to \mathbf{x}') = q(\mathbf{x}' \mid \mathbf{x})\alpha(\mathbf{x}, \mathbf{x}')$, and detailed balance asks $\pi(\mathbf{x})q(\mathbf{x}' \mid \mathbf{x})\alpha(\mathbf{x}, \mathbf{x}') = \pi(\mathbf{x}')q(\mathbf{x} \mid \mathbf{x}')\alpha(\mathbf{x}', \mathbf{x})$. So $\alpha(\mathbf{x}, \mathbf{x}')/\alpha(\mathbf{x}', \mathbf{x})$ must equal the ratio r below; the largest choice with both values in $[0, 1]$ sets the larger of the two to 1:

$$\alpha(\mathbf{x}, \mathbf{x}') = \min\Big\{1,\ r\Big\}, \qquad r = \frac{\tilde{\pi}(\mathbf{x}')\, q(\mathbf{x} \mid \mathbf{x}')}{\tilde{\pi}(\mathbf{x})\, q(\mathbf{x}' \mid \mathbf{x})}. \tag{11.8}$$

The normalising constant of π cancels. For a symmetric proposal (random walk $\mathbf{x}' = \mathbf{x} + \varepsilon$) the Hastings factor $q(\mathbf{x} \mid \mathbf{x}')/q(\mathbf{x}' \mid \mathbf{x})$ is 1: always accept uphill moves, accept downhill moves with probability $\tilde{\pi}(\mathbf{x}')/\tilde{\pi}(\mathbf{x})$.

Proposition 11.4 (Gibbs is MH with acceptance one)**.** *Resampling coordinate i from its full conditional is an MH step with proposal $q(\mathbf{x}' \mid \mathbf{x}) = \pi(x_i' \mid \mathbf{x}_{-i})\mathbb{1}\{\mathbf{x}'_{-i} = \mathbf{x}_{-i}\}$, and its acceptance ratio is identically* 1.

Proof. With $\mathbf{x}'_{-i} = \mathbf{x}_{-i}$ and $\pi(\mathbf{x}) = \pi(x_i \mid \mathbf{x}_{-i})\pi(\mathbf{x}_{-i})$,

$$r = \frac{\pi(x_i' \mid \mathbf{x}_{-i})\pi(\mathbf{x}_{-i}) \cdot \pi(x_i \mid \mathbf{x}_{-i})}{\pi(x_i \mid \mathbf{x}_{-i})\pi(\mathbf{x}_{-i}) \cdot \pi(x_i' \mid \mathbf{x}_{-i})} = 1. \qquad \square$$

Random-walk MH with step size s: propose $y = x^{(t)} + s\,\varepsilon$, $\varepsilon \sim N(0,1)$, draw $u \sim U(0,1)$, and set $x^{(t+1)} = y$ if $\log u < \log \tilde{\pi}(y) - \log \tilde{\pi}(x^{(t)})$ (work in logs), else $x^{(t+1)} = x^{(t)}$.

Worked Example 11.8 (One Metropolis–Hastings step by hand)**.** Target Gamma$(3, 1)$, $\tilde{\pi}(x) = x^2 e^{-x}$ $(x > 0$; the constant $\frac{1}{2}$ cancels). The chain is at $x = 2$ and the proposal is

$y = 3.5$. Compute the acceptance probability (a) for a symmetric random-walk proposal, (b) for the independence proposal $q = \text{Exp}(\text{mean } 3)$, $q(z) = \frac{1}{3}e^{-z/3}$, which ignores the current state.

Solution. (a) The Hastings factor is 1, so $r = \dfrac{3.5^2 e^{-3.5}}{2^2 e^{-2}} = \dfrac{12.25}{4} e^{-1.5} = 3.0625 \cdot 0.2231 = 0.6833$ and $\alpha = \min\{1, 0.6833\} = 0.6833$. (b) Now $q(\mathbf{x} \mid \mathbf{x}') = q(2)$ and $q(\mathbf{x}' \mid \mathbf{x}) = q(3.5)$, so the Hastings factor is $q(2)/q(3.5) = e^{(3.5-2)/3} = e^{0.5} = 1.6487$, $r = 0.6833 \cdot 1.6487 = 1.1266$ and $\alpha = 1$: the move is accepted for sure, because the proposal is less likely to suggest 3.5 than 2.

PYTHON LAB · `s11_examples.py` · `ex_mh_acceptance()`

```
random walk:  ratio = 0.6833, accept prob = 0.6833 | independence:  Hastings
factor 1.6487, ratio 1.1266, accept 1.0000
```

PART III · MCMC AND BEYOND

11.10 Is the Chain Done? Convergence Diagnostics

MCMC output is a correlated sequence, and no diagnostic can prove convergence; they can only reveal failure.

Definition 11.4 (Autocorrelation time, ESS, $\hat{R}$)**.** For a stationary chain with lag-k autocorrelation ρ_k of $f(\mathbf{x}^{(t)})$, the *integrated autocorrelation time* is $\tau = 1 + 2\sum_{k \geq 1} \rho_k$ and $\text{ESS} = N/\tau$. For M chains of length n with within-chain variance W (mean of the M chain variances, each with divisor $n-1$) and between-chain variance $B = n \cdot \text{Var}(\text{chain means})$ (divisor $M-1$), the Gelman–Rubin statistic is

$$\hat{R} = \sqrt{\widehat{\text{var}}^{+}/W}, \qquad \widehat{\text{var}}^{+} = \tfrac{n-1}{n} W + \tfrac{1}{n} B. \tag{11.9}$$

Here $\widehat{\text{var}}^{+}$ estimates the target variance from all chains together; if the chains have not mixed, the between-chain spread B inflates it relative to W and $\hat{R} > 1$. In practice the sum defining τ is truncated at the first lag where the (estimated) autocorrelations become negative (Geyer's initial-positive-sequence rule, used by `s11_examples.py`).

In practice: inspect trace plots for trends, stuck chains and rare mode switches; discard a burn-in; thin only to save memory (thinning never increases the ESS of all draws); report the ESS, not N (an ESS of a few hundred suffices for two-digit summaries); and run several dispersed chains, requiring $\hat{R} < 1.01$ — values above 1.1 mean the chains disagree.

PART III · MCMC AND BEYOND

11.11 Variational Inference in One Section

Variational inference (VI) replaces sampling by optimisation. For latent $\mathbf{z}$, data $\mathbf{x}$ and any density $q(\mathbf{z})$,

$$\log p(\mathbf{x}) = \underbrace{\mathbb{E}_q[\log p(\mathbf{x}, \mathbf{z})] - \mathbb{E}_q[\log q(\mathbf{z})]}_{\text{ELBO}(q)} + \text{KL}\big(q(\mathbf{z}) \,\|\, p(\mathbf{z} \mid \mathbf{x})\big), \tag{11.10}$$

so maximising the ELBO over a tractable family is the same as minimising $\mathrm{KL}(q\|p(\cdot \mid \mathbf{x}))$; the ELBO is the bound EM maximised in Session 10. In *mean field*, $q(\mathbf{z}) = \prod_j q_j(z_j)$, and coordinate ascent (CAVI) iterates

$$\log q_j^*(z_j) = \mathbb{E}_{q_{-j}}\big[\log p(\mathbf{x}, \mathbf{z})\big] + \text{const}, \tag{11.11}$$

a "Gibbs sampler with expectations in place of samples" that needs only each variable's Markov blanket. For a bivariate Gaussian target $N(\boldsymbol{\mu}, \Sigma)$ with precision $\Lambda = \Sigma^{-1}$, $\log p = -\frac{1}{2}\Lambda_{11}(z_1 - \mu_1)^2 - \Lambda_{12}(z_1 - \mu_1)(z_2 - \mu_2) + (\text{terms without } z_1)$; taking $\mathbb{E}_{q_2}$ replaces z_2 by its mean m_2, and completing the square in z_1, (11.11) gives $q_1 = N(m_1, 1/\Lambda_{11})$ with $m_1 = \mu_1 - \Lambda_{12}(m_2 - \mu_2)/\Lambda_{11}$ (and symmetrically for q_2).

Example. Target $N\big((1, -1), \Sigma\big)$ with unit variances and correlation $\rho = 0.9$. Then $\Lambda = \frac{1}{1-\rho^2}\begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$, so $\Lambda_{11} = \Lambda_{22} = 1/(1 - 0.81) = 1/0.19 = 5.26$ and $-\Lambda_{12}/\Lambda_{11} = \rho = 0.9$. Each factor has variance $1/5.26 = 0.19$ against a true marginal variance of 1 (mean-field factors have the conditional variance $1/\Lambda_{ii}$, which is smaller than the marginal variance). At the optimum the means are exact, and the Gaussian KL formula gives $\mathrm{KL}(q^*\|p) = \frac{1}{2}\big[\mathrm{tr}(\Lambda S_q) - 2 + \ln(\det\Sigma/\det S_q)\big]$ with $S_q = \mathrm{diag}(0.19, 0.19)$: $\mathrm{tr}(\Lambda S_q) = 2$ and $\det\Sigma/\det S_q = (1-\rho^2)/(1-\rho^2)^2$, so $\mathrm{KL}(q^*\|p) = -\frac{1}{2}\ln(1-\rho^2) = -\frac{1}{2}\ln 0.19 = 0.8304$. CAVI from $(0, 0)$ updates $m_1 = 1 + 0.9(m_2 + 1)$, then $m_2 = -1 + 0.9(m_1 - 1)$: sweep 1 gives $m_1 = 1 + 0.9 \cdot 1 = 1.90$ and $m_2 = -1 + 0.9 \cdot 0.90 = -0.19$; then $(1.73, -0.34)$, …, $(1.39, -0.65)$ after five sweeps. The means converge to $(1, -1)$, but the error $m_1 - 1$ only shrinks by the factor $\rho^2 = 0.81$ per sweep $(0.90, 0.729, \dots)$ — the same zig-zag as Gibbs.

Which KL? Fit a single Gaussian $q = N(m, s^2)$ to the bimodal mixture $\pi = 0.3\,N(-3, 0.8^2) + 0.7\,N(2, 1)$ (mean $0.3(-3) + 0.7(2) = 0.5$). Minimising $\mathrm{KL}(p\|q)$ matches moments: $m = 0.5$ and $s^2 = 0.3(0.64 + 9) + 0.7(1 + 4) - 0.5^2 = 6.142$, $s = 2.478$, a broad q covering both modes. Minimising $\mathrm{KL}(q\|p)$ (the VI objective; numerical grid search over m, s in steps of 0.05) gives $m = 1.950$, $s = 1.050$ with $\mathrm{KL} = 0.350$: q locks onto the heavier mode. For comparison, $q = N(2, 1)$ exactly would give $\mathrm{KL}(q\|p) \approx -\ln 0.7 = 0.357$, because under that q the target density is about $0.7\,q$.

PYTHON LAB · s11_examples.py · ex_mean_field_gaussian(), ex_kl_asymmetry()

```
mean-field var 0.19 (true 1); KL(q*||p) = 0.8304; sweep 1:  (1.9000,
-0.1900); sweep 5:  (1.3874, -0.6513)
KL(p||q) fit:  mean 0.500, sd 2.478 (var 6.142) | KL(q||p) fit:  mean 1.950,
sd 1.050, KL 0.350; -ln 0.7 = 0.357
```

AI CONNECTION · Generative Models

A variational autoencoder (Kingma & Welling 2014; course session 4) maximises the ELBO (11.10) with an *amortised* $q_\phi(\mathbf{z} \mid \mathbf{x})$: an encoder outputs the variational parameters for every $\mathbf{x}$ instead of running CAVI per data point; diffusion models also train on an ELBO. $\mathrm{KL}(q\|p)$ is infinite unless $q = 0$ wherever $p = 0$ (zero-forcing, mode-seeking), whereas $\mathrm{KL}(p\|q)$ punishes missing mass and matches moments: hence over-confident VI and VAE posteriors.

PART III · MCMC AND BEYOND

11.12 Particle Filtering

For dynamic models (HMMs and dynamic BNs with continuous or huge state spaces), the filtering recursion of Session 7, $b_t(x_t) \propto p(o_t \mid x_t) \int p(x_t \mid x_{t-1}) b_{t-1}(x_{t-1})\, dx_{t-1}$, has no closed form outside

the linear-Gaussian (Kalman) case. A *particle filter* (bootstrap filter; Gordon, Salmond & Smith 1993) represents b_t by N weighted samples and runs sequential importance sampling with the transition model as proposal: 1 *propagate* $x_t^{(k)} \sim p(x_t \mid x_{t-1}^{(k)})$; 2 *weight* $w_t^{(k)} \propto w_{t-1}^{(k)} p(o_t \mid x_t^{(k)})$ and normalise; 3 *resample* N particles with probabilities $w_t^{(k)}$ (e.g. systematic resampling) when the ESS (11.7) drops below $N/2$, and reset all weights to $1/N$. Without step 3 the weights degenerate: after a few steps one particle carries almost all the weight.

PART IV

Sampling and Structure in Agents

Rollouts, beliefs, votes, world models and interaction graphs

PART IV · AI CONNECTIONS

11.13 Sampling Inside Agents

AI CONNECTION · Reinforcement Learning

Monte Carlo policy evaluation estimates $V^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$ by averaging sampled returns. On the 5-state random walk of Sutton & Barto (2018, Example 6.2) — states $A, \ldots, E$, start in C, step left or right with probability $\frac{1}{2}$, reward 1 for exiting on the right and 0 otherwise, no discounting — $V(C) = 0.5$. 100 simulated episodes (seed 2028) give $\hat{V}(C) = 0.550$, SE $= s/\sqrt{100} = 0.050$, 95% CI $0.550 \pm 1.96 \cdot 0.050 = [0.452, 0.648]$, mean episode length 8.50 (theory $3 \cdot 3 = 9$, the gambler's-ruin duration from distance 3 to each exit); SE $= 0.01$ would need $\sigma^2/0.01^2 = 0.25/0.0001 = 2500$ episodes, the variance problem that temporal-difference learning attacks by bootstrapping (`ex_mc_policy_eval()`).

AI CONNECTION · Agentic AI

Self-consistency (Wang et al. 2023) samples N chain-of-thought paths and returns the plurality answer, a Monte Carlo estimate of the mode of the answer distribution. Simulation (200,000 hypothetical questions per N, ties broken at random): for a model right 40% of the time with errors spread over four wrong answers (15% each), the accuracy of the plurality vote rises from 0.40 ($N = 1$) to 0.55 ($N = 5$), 0.71 ($N = 11$) and 0.96 ($N = 41$). If one wrong answer is more likely than the right one (45%, the other three 5% each), voting *converges to the wrong answer*: 0.36 at $N = 41$. Sampling reduces variance, never bias; best-of-N and tree-of-thought rollouts (course session 26) share this caveat (`ex_self_consistency()`).

AI CONNECTION · Multi-Agent AI

Five agents log continuous action signals (simulation, seed 2026, $n = 2000$ hypothetical rounds of the linear-Gaussian model $A_1 = \varepsilon_1$, $A_2 = 0.8A_1 + 0.6\varepsilon_2$, $A_3 = 0.7A_2 + 0.7\varepsilon_3$, $A_4 = 0.6A_1 + 0.8\varepsilon_4$, $A_5 = \varepsilon_5$ with independent standard normal ε_j): A_1 influences A_2 and A_4, A_2 influences A_3, A_5 acts alone. Marginal correlations link almost everyone: $\mathrm{corr}(A_3, A_4) = 0.35$. Partial correlations given the other three agents (Session 8: zeros of the precision matrix) with Fisher-z tests at level 0.01 ($|S| = 3$, so $z = \frac{1}{2} \ln \frac{1+\hat{r}}{1-\hat{r}} \sqrt{2000 - 6}$)

recover exactly the interaction graph $A_1 - A_2$, $A_1 - A_4$, $A_2 - A_3$; the partial correlation of A_3, A_4 is 0.00. As in Worked Example 11.3, the graph is undirected where observational data cannot tell leaders from followers (`ex_multiagent_graph()`).

PART IV · AI CONNECTIONS

11.14 Summary

Table 11.1. Cheat sheet of key formulas.

Item	Formula
Likelihood of a DAG	$\ell = N\sum_i \hat{I}(X_i;U_i) - N\sum_i \hat{H}(X_i)$; AIC $= \ell - d$, BIC $= \ell - \frac{d}{2}\log N$
Bayesian score	$\prod_{i,u} \frac{\Gamma(\alpha_u)}{\Gamma(\alpha_u+N_u)} \prod_x \frac{\Gamma(\alpha_{x\mid u}+N_{x,u})}{\Gamma(\alpha_{x\mid u})}$
Monte Carlo	$\hat{\mu} = \frac{1}{N}\sum f(x^{(m)})$, SE $= \sigma/\sqrt{N}$; CI $\hat{\mu} \pm 1.96\, s/\sqrt{N}$
Importance sampling	$w = p/q$; $q^* \propto \lvert f\rvert p$; ESS $= (\sum w)^2/\sum w^2$
Metropolis–Hastings	$\alpha = \min\{1, \tilde{\pi}(x')q(x \mid x')/[\tilde{\pi}(x)q(x' \mid x)]\}$
MCMC error	ESS $= N/\tau$, $\tau = 1 + 2\sum_k \rho_k$; $\hat{R} = \sqrt{((n-1)W/n + B/n)/W}$
VI	$\log p(x) = \mathrm{ELBO}(q) + \mathrm{KL}(q\Vert p(\cdot \mid x))$; $\log q_j^* = \mathbb{E}_{-j} \log p(x,z) + c$

THINK ABOUT IT (1) Two DAGs differing in one edge direction get the same BIC. What does that say about using a learned structure to predict interventions? (2) All chains report $\hat{R} = 1.00$, yet no trace ever switches between two known modes. Converged?

Further reading. Koller & Friedman (2009), Chs. 12, 18 and 19; Robert & Casella (2004), *Monte Carlo Statistical Methods*; Spirtes, Glymour & Scheines (2000), *Causation, Prediction, and Search*.

CONNECTIONS AND TAKEAWAYS

We now have the probabilistic toolkit: represent independence with a graph (Sessions 6, 8), compute exactly when the treewidth allows (Session 7), learn parameters (Session 10) and structure (today), and fall back to sampling or variational approximations when exact computation is out of reach. All of it describes a single joint distribution P obtained by *observing* the world. The question none of these tools can answer is what happens to Y if we *change* the world — give every patient the drug rather than observe who took it. That is not $P(Y \mid X = x)$, and **Session 12** shows with real numbers that confusing the two can even reverse the sign of an effect.